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Your grade: 93.33%

Your score: 93.33% · Your high score: 100%

Module 3 Summative Assessment

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Review Learning Objectives

Instructions

1 / 1 point

1. What is the scaled inverse- χ^2 distribution used for in Bayesian analysis of normal data?

☐ To model the mean of the data

☒ To model the variance of the data

☐ To model the joint prior distribution

☐ To model the posterior distribution

Correct

Due

Attempts

The scaled inverse- χ^2 distribution is used to model the variance of normal data in the Bayesian analysis of normal data.

Submitted

Nov 2, 11:46 PM EDT

Retry

2. Why is the posterior predictive distribution useful?

☐ It records for parameter values at least 80%. We keep your highest score.

☒ It allows for predictions.

☐ It integrates out the data

☐ It simplifies the model

Correct

View submission

See feedback

The posterior distribution allows us to compute probabilities for predictions.

3. Suppose two instructors are being considered for one position. In the last class Instructor A taught, 15 out of the 19 students approved of the way the instructor managed the class. In the last class Instructor B taught, 20 out of the 27 students approved of the way the instructor managed the class. Using Bayesian inference, construct a 95% Credibility Interval of the difference between the two rates. Based on the data, what would you conclude?

☐ The instructors are different since the credibility interval includes 0.

☐ The instructors are equivalent since the credibility interval does not include 0.

☐ The instructors are equivalent since the credibility interval does not include 5.

☒ The instructors are equivalent since the credibility interval includes 0.

Correct

Correct: The credibility interval is roughly symmetric about 0.

4. In the context of normal data with a noninformative prior distribution, what is the joint prior density for μ and σ^2 ?

☒ Proportional to $(\sigma^2)^{-1}$

☐ Uniform on (μ, σ^2)

☐ Proportional to (μ, σ^2)

☐ Uniform on $(\mu, \log \sigma^2)$

Correct

Correct: For normal data with a noninformative prior, the joint prior density for μ and σ^2 is proportional to $(\sigma^2)^{-1}$.

5. What is the posterior predictive solution?

☐ The calculation of the posterior distribution of parameters.

☐ The estimation of prior probabilities.

☐ The simulation of the posterior distribution.

☒ The computation of the likelihood of unseen data.

Correct

Correct: The posterior predictive solution computes the likelihood of some unseen y slide given the available data, y .

6. Which is a common alternative to histograms when studying two parameters.

☐ Graphs

☐ Line plots

☒ Box plots

☐ Contour plots

Incorrect

Box plots are inappropriate for the data described

7. What type of prior is considered weakly informative?

☐ Prior that dominates the likelihood

☐ Noninformative prior

☐ Prior with high variance

☒ A prior with little information

Correct

Correct: A weakly informative prior is one that uses some information; it is a mixture of an informative prior and a noninformative one.

8. In Bayesian analysis, what is the purpose of using a noninformative prior?

☒ To reflect a state of prior ignorance

☐ To simplify computations

☐ To incorporate strong prior beliefs

☐ None of the above

Correct

Correct: Noninformative priors are used to reflect a state of prior ignorance and to let the data drive the posterior distribution.

9. How should you sample from the conjugate posterior distribution for data normally distributed with unknown mean and unknown variance?

☐ Sample the mean first and then the variance

☐ It is not possible to sample these parameters from the posterior

☐ Take both at the same time

☒ Sample first the variance then the mean

Correct

Correct: It is easiest to sample the variance from the inverse chi squared distribution based on the data and then sample the mean from the marginal posterior distribution.

10. Before and after a major policy announcement, an independent polling organization conducted surveys of 800 registered voters. The results are summarized as follows: Pre-Policy Announcement Survey:

- Voters in-favor: 320
- Voters against: 380
- Undecided/Other: 50

Post-Policy Announcement Survey:

- Voters in-favor: 400
- Voters against: 350
- Undecided/Other: 50

What is the posterior probability that there was a shift toward being in favor?

☒ 99%

☐ 61%

☐ 74%

☐ 48%

Correct

Correct: When you run the simulation, you should find an estimate of roughly 99%.

11. What is the purpose of using the factorization $p(\theta|y) = \int p(\theta|y_1, y_2)p(y_2|y_1)\theta dy_2$

☐ To explicitly evaluate this integral to compute $p(\theta|y)$

☐ To derive the likelihood function

☐ To simplify the computation

☒ To show that $p(\theta|y)$ is a mixture of conditional posterior distributions

Correct

Correct

12. Which distribution is used to model covariance matrices?

☒ Inverse-Wishart

☐ Normal

☐ t-distribution

☐ Scaled Inverse- χ^2

Correct

Correct: The inverse-Wishart distribution is used to model covariance matrices in the scaled inverse-Wishart model.

13. In a biology class, two instructors (Instructor A and Instructor B) are evaluated based on their recent classes. Instructor A received approval from 15 out of 19 students in the first class and 20 out of 27 students in the second class. Using Bayesian inference to estimate the approval rates, which of the following statements about Bayesian analysis in this context is correct?

☐ The Bayesian approach assumes a normal distribution for the approval rates of Instructor A and Instructor B.

☒ The prior distributions for the approval rates of Instructor A and Instructor B are modeled using Beta distributions.

☐ The credibility interval for the difference in approval rates between Instructor A's two classes can be directly calculated using frequentist methods.

☐ The posterior distribution for the approval rates of Instructor A and Instructor B will be uniform distributions.

Correct

Correct: Because the responses are binary, our (weakly informative) priors are generally Beta distributions.

14. What is the primary reason to simulate the posterior distribution for multi-parameter models?

☐ Simulating the posterior distribution allows us to visualize the uncertainty in each parameter individually.

☒ Multi-parameter models often have complex interactions between parameters, which can only be captured and understood through simulation.

☐ It helps in computing the prior distribution of parameters efficiently.

☐ Simulation is a faster method compared to traditional inference techniques for multi-parameter models.

Correct

Correct: For many cases, we cannot solve them analytically.

15. How is the marginal posterior distribution of a parameter of interest obtained?

☐ By ignoring all nuisance parameters

☒ By integrating the joint posterior distribution over nuisance parameters

☐ By setting all other parameters to zero

☐ By using a noninformative prior

Correct

Correct: The marginal posterior distribution is obtained by integrating the joint posterior distribution over the nuisance parameters.